Restricted progressions after intersection with the dual recurrence filter

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Abstract

We construct a set $D \subseteq \mathbb{N}$ which remains 2-large after intersection with every member of the dual filter of single topological recurrence, although a three-coloring has no monochromatic 600-term progression of difference in D. In fact, for each k we can force a k-monochromatic progression and agreement of an arbitrary finite auxiliary coloring on its first two terms, while blocking progressions of one fixed length in $k+1$ colors. The main ingredient is a finite graph with large chromatic number whose edge differences all support the required configurations on a nilfactor. Its vertex phases are obtained by correcting a shift-graph construction of Green inside a torus fiber of bounded codimension. A summable sequence of frequencies lets all finite stages share one obstructing coloring.

1 The question and the stronger statement

Throughout, $\mathbb{N} = \{1, 2, \ldots\}$ and $[r] = \{1, \ldots, r\}$. A set D is k-large if every k-coloring of $\mathbb{N}$ admits monochromatic arithmetic progressions of arbitrary finite length with positive common difference in D. It is large if this holds for all finite numbers of colors. Write $\mathcal{P}$ for the family of sets of single topological recurrence. Its dual $\mathcal{P}^*$ consists of the supersets of

$$N^1(U) = \{d \geq 1 : U \cap T^{-d}U \neq \varnothing\}$$

for nonempty open U in minimal compact metric systems (X, T) with T a homeomorphism. The question is whether 2-largeness of $D \cap A$ for every $A \in \mathcal{P}^*$ forces D to be large.

For a niltranslation T on a compact nilmanifold $Y = G/\Gamma$, a point y, and an open neighborhood V of y, put $N(y, V) = \{n \geq 1 : T^n y \in V\}$. We use presentations by a nilpotent Lie group with a discrete cocompact subgroup, allowing disconnected systems.

Definition 1.1. Say that D has W_s if the following holds for every such (Y, T, y, V), every $h \geq 2$, every $c : N(y, V) \to [s]$, every $r \geq 1$, and every $g : \mathbb{N} \to [r]$. There exist $a, d \geq 1$ with $d \in D$ such that

$$\begin{aligned} &a, a+d, \ldots, a+(h-1)d \in N(y, V), \\ &c(a) = c(a+d) = \cdots = c(a+(h-1)d), \\ &g(a) = g(a+d). \end{aligned}$$

Thus the auxiliary pair occurs at the first two sites of the same progression, regardless of the number of auxiliary colors.

Theorem 1.2. *For every $k \geq 1$ there is a set $D_k \subseteq \mathbb{N}$ with W_k and a $(k+1)$-coloring of $\mathbb{N}$ with no monochromatic D_k-progression of length $200(k+1)$.*

Corollary 1.3. *There is a set D which is not 3-large, although $D \cap A$ is 2-large for every $A \in \mathcal{P}^*$.*

Proof. Take $D = D_2$. If $A \supseteq N^1(U)$, choose a finite cover $X = \bigcup_{i=1}^{r} T^{-j_i}U$ by minimality. Fix $z \in X$ and choose $g(n) = i$ with $T^{n+j_i}z \in U$. Whenever $g(a) = g(a+d) = i$, both $T^{a+j_i}z$ and $T^{a+d+j_i}z$ lie in U, so $d \in A$. Apply W_2 with this auxiliary coloring, the one-point nilsystem, and the given binary coloring. The resulting progression has difference in $D \cap A$. The obstructing three-coloring is furnished by Theorem 1.2. □

Where the auxiliary coloring enters. Knowing that D is both topologically recurrent and 2-large is not enough to identify a difference with both properties. Our construction first arranges many possible differences, indexed by the edges of a graph. Every edge already satisfies the geometric constraints. Coloring the vertices by auxiliary labels leaves a monochromatic edge, and polynomial recurrence lifts the configuration for that edge. Only the dimension required to obtain this graph depends on r; the obstruction length does not.

2 The finite-stage sets

In the induction from s to $s+1$, let B have W_s and let $f : \mathbb{N} \to \{0, \dots, s\}$ block every monochromatic B-progression of length H. The first step instead takes $s = 0$, $B = \varnothing$, $H = 2$, and constant f; no zero-color recurrence hypothesis is used. Set

$$u = s+1, \quad q = u+1, \quad Q = 100q, \quad \delta = 1/Q, \quad \eta = (10^6 Q^2)^{-1}, \quad \tau = (4\pi q H)^{-1}. \tag{2.1}$$

For a finite list of positive frequencies $\alpha_1, \dots, \alpha_M$ with $1, \alpha_1, \dots, \alpha_M$ rationally independent, write $\alpha = (\alpha_1, \dots, \alpha_M)$. Use centered coordinate representatives $\theta_j \in [-1/2, 1/2)$ on $\mathbb{T}^M$. The open set O_M consists of the displacements for which some i satisfies

$$|\theta_i + \delta| < \eta, \qquad |\theta_j| < \eta \quad (j \neq i), \qquad \sum_{j \neq i} |\theta_j| < 2\delta. \tag{2.2}$$

Put

$$G_M = \{d \geq 1 : d\alpha \in O_M\}, \qquad B_M = \{d \in B : \textstyle\sum_{j \leq M} \|d\alpha_j\|_{\mathbb{R}/\mathbb{Z}} < \tau\}, \qquad S_M = G_M \cup B_M. \tag{2.3}$$

Here $\|t\|_{\mathbb{R}/\mathbb{Z}}$ denotes distance to the nearest integer. The G_M directions will handle the case where one primary color covers a factor window. The B_M directions preserve the previous inductive conclusion when a primary color is absent above a smaller factor window.

3 A shift graph of admissible differences

The shift graph and the ramp vectors used below come from Green's proof of single recurrence [3, Appendix A]. Here we put corrected ramp vectors in a torus subspace determined by the prescribed nilsystem, and choose the vertex times so that each edge gives an entire progression in the prescribed window.

Lemma 3.1. *Fix $h \geq 2$, a dimension bound $C \geq 0$, a number $r \geq 1$ of labels, and the palette parameter $q \geq 2$. Use $Q, \delta = 1/Q, \eta, O_M$ and G_M as above. For every sufficiently large M, depending only on h, C, r, q, the following holds. Let (Y, T) be a nilsystem of dimension at most C, let $y \in Y$, and let V be an open neighborhood of y. There is a finite directed graph with distinct positive integer times n_u at its vertices such that $T^{n_u} y \in V$ for every vertex u and, for every directed edge $u \to v$,*

$$d_{uv} = n_v - n_u > 0, \qquad d_{uv} \in G_M, \qquad T^{n_u + j d_{uv}} y \in V \quad (0 \leq j < h).$$

Every labeling of the vertices by at most r labels gives an edge whose endpoints have the same label.

Proof. The torus directions available above the starting point. Put $J = \{1, \dots, h-1\} \cup \{-1, \dots, -(h-2)\}$, with the negative part empty if $h = 2$. On $Y^J \times \mathbb{T}^M$ take the translation whose coordinates are T^j for $j \in J$ and the rotation $t \mapsto t + \alpha$ on the torus. Let K be the orbit closure of $w = ((y)_{j \in J}, 0)$ and let K_0 be its connected component through w. By the orbit-closure theorem for niltranslations, K is minimal, its finitely many connected components are cyclically permuted, and $K_0 = L_0 w$ for a connected closed subgroup L_0 of the ambient product group [4, Theorem B and Sections 2.14–2.16]. The projection of K_0 onto the torus is the image of L_0 under the torus homomorphism, so it is a connected subgroup and is closed by compactness of K_0. The full projection of K is $\mathbb{T}^M$, by rational independence of $1, \alpha_1, \dots, \alpha_M$. Since finitely many translates of a proper connected subtorus cannot cover $\mathbb{T}^M$, the torus projection of L_0 is surjective. Its derivative is therefore surjective as well.

Present Y as G/Γ with Γ discrete, so $\dim G = \dim Y \leq C$, also when G is disconnected. The Lie algebra of L_0 has dimension at least M, while its projection onto the Lie algebras of the Y-coordinates has rank at most $b = (2h-3)C$. Consequently the real subspace

$$H = \{v \in \mathbb{R}^M : (0, \dots, 0, v) \in \mathrm{Lie}(L_0)\}$$

has codimension at most b. Exponentiating these Lie algebra elements gives

$$((y)_{j \in J}, v \bmod \mathbb{Z}^M) \in K_0 \qquad (v \in H).$$

A shift graph inside the fiber. Set $L = \lceil 8\delta/\eta \rceil$ and choose $m = R_r^{(L)}(L+1)$, the finite Ramsey number for r-labelings of L-element subsets. Let $A : \mathbb{R}^M \to E = \mathbb{R}^M/H$ be the quotient map and $v_i = Ae_i$ the images of the standard basis vectors. Give E the quotient ℓ^1 norm, whose unit ball is the convex hull of the vectors $\pm v_i$. For $\varepsilon = \eta/(128\delta(L+1))$, that ball is covered by at most $(1 + 2/\varepsilon)^{\dim E}$ balls of radius ε. Indeed, the balls of radius $\varepsilon/2$ about a maximal ε-separated set have disjoint interiors and lie in the ball of radius $1 + \varepsilon/2$, so the bound follows by comparing volumes. In dimension zero one ball suffices. Thus, if $M \geq m\lceil (1 + 2/\varepsilon)^b \rceil$, there is a set I of m distinct indices for which the v_i, $i \in I$, lie in one ball with center z.

Take as vertices the increasing L-tuples from I, with a directed edge from $(i_1, \dots, i_L)$ to $(i_2, \dots, i_{L+1})$ whenever $i_1 < \cdots < i_{L+1}$. Any r-labeling of these vertices has a homogeneous $(L+1)$-element subset by the choice of m; its first and last L-subsets give the desired equally labeled edge. For a vertex $u = (i_1, \dots, i_L)$ define

$$x_u = -\sum_{\ell=1}^{L} \frac{\delta \ell}{L} e_{i_\ell}, \qquad \mu = \sum_{\ell=1}^{L} \frac{\delta \ell}{L} = \frac{\delta(L+1)}{2}.$$

The quotient vector Ax_u is within $\mu\varepsilon$ of $-\mu z$. Fix one reference vertex u_*; then $A(x_u - x_{u_*})$ has quotient norm at most $2\mu\varepsilon = \eta/128$. By the definition of quotient norm, choose $c_u \in \mathbb{R}^M$ with $Ac_u = A(x_u - x_{u_*})$ and $\|c_u\|_1 < \eta/64$. The corrected vector $p_u = x_u - x_{u_*} - c_u$ belongs to H. For an edge $u \to v$, the uncorrected difference $x_v - x_u$ has one coordinate $-\delta$ at i_{L+1}, has value δ/L at

$i_1, \dots, i_L$, and vanishes elsewhere. The correction changes this difference in ℓ^1 norm by less than $\eta/32$.

Realizing the vertices at positive times. Equip $\mathbb{T}^M$ with the distance $d_1(t,t') = \sum_i \|t_i - t'_i\|_{\mathbb{R}/\mathbb{Z}}$ and write $[p_u] = p_u \bmod \mathbb{Z}^M$. Process the vertices in lexicographic order, in which every directed edge goes from an earlier to a later vertex. Choose n_v larger than all the previously chosen times and positive, subject to

$$\begin{aligned} &d_1(n_v\alpha, [p_v]) < \eta/32, \\ &T^{-tn_v} y \in V \qquad\qquad (1 \le t \le h-2), \\ &T^{jn_v} y \in V \cap \bigcap_{u \text{ earlier}} T^{(j-1)n_u} V (1 \le j < h). \end{aligned}$$

Each target for a Y-coordinate is a neighborhood of y, because the earlier choices gave $T^{-(j-1)n_u} y \in V$ for $j \ge 2$, and $y \in V$ for $j = 1$. Together with the phase condition these targets contain the point $((y)_{j\in J}, [p_v])$ of K_0. Every positive orbit tail of w is dense in K, so an arbitrarily late admissible n_v exists. In particular $T^{n_v} y \in V$ at every vertex.

For an edge $u \to v$ the construction gives $d_{uv} > 0$ and, for $1 \le j < h$,

$$T^{n_u + j d_{uv}} y = T^{(1-j)n_u} T^{jn_v} y \in V.$$

Finally, $d_{uv}\alpha$ lies within $3\eta/32$ in torus ℓ^1 distance of $[x_v - x_u]$, by the two vertex approximations and the correction estimate. Its distinguished centered coordinate therefore differs from $-\delta$ by less than η, each other coordinate has absolute value at most $\delta/L + 3\eta/32 < \eta$, and their total absolute value is at most $\delta + 3\eta/32 < 2\delta$. Here the distinguished coordinate stays away from the endpoint of the centered fundamental interval since $\delta + \eta < 1/2$. Thus $d_{uv}\alpha \in O_M$, so $d_{uv} \in G_M$ as claimed. □

4 Lifting an edge with matching auxiliary colors

For a minimal system X, let $\pi : X \to Z$ be its maximal infinite-step pronilfactor. Thus Z is an inverse limit of finite-dimensional minimal nilsystems, and every minimal nilsystem factor of X factors through Z. We first state the exact consequence of Qiu's theorem used below.

Lemma 4.1 (Positive lifting)**.** *Suppose $U_0, \dots, U_{h-1}$ are open in X and, for some $m \ge 1$,*

$$\bigcap_{j=0}^{h-1} T^{-jm} \operatorname{Int} \pi(U_j) \neq \varnothing.$$

There are $x \in X$ and $n \in \mathbb{Z}$ such that $T^{j(m+n^2)} x \in U_j$ for $0 \le j < h$.

Proof. Theorem B of Qiu [6] supplies an almost one-to-one modification of the factor diagram: there are surjective factor maps $\tau_X : X^* \to X$, $\sigma : Z^* \to Z$, and $\pi^* : X^* \to Z^*$ with $\sigma\pi^* = \pi\tau_X$, where σ is almost one-to-one. Choose a point z of the displayed open intersection with singleton σ-fiber. For $V_j^* = \tau_X^{-1}(T^{-jm} U_j)$, the sets $\pi^*(V_j^*)$ all contain that unique point above z. Indeed, choose a point of $T^{-jm}U_j$ above z and lift it by τ_X; its π^*-image must be in the singleton fiber. Apply Qiu's theorem to $V_0^*, \dots, V_{h-1}^*$ and the distinct polynomials $p_j(n) = jn^2$, $1 \le j < h$. Projecting the resulting configuration by τ_X gives the assertion. In particular, $m + n^2 > 0$ even though n may be negative. □

Proposition 4.2 (Finite-stage recurrence)**.** *Fix $h \geq 2$, a dimension bound C, and an auxiliary color bound r. For every sufficiently large M, depending only on h, C, r, q, the set S_M of (2.3) satisfies the conclusion of W_{s+1} on all windows in nilsystems of dimension at most C, for this h and at most r auxiliary colors. This holds for every choice of rationally independent frequencies as specified above.*

Proof. Let (Y, T, y, V) be a test of dimension at most C. Extend its primary coloring c arbitrarily to a two-sided sequence, and likewise extend the auxiliary coloring g. Let X be a minimal subsystem of the forward limit set of the point $(c, g, y, 0)$, with shifts on the two symbolic coordinates, T on Y, and addition of α on $\mathbb{T}^M$. The joint orbit closure of $(y, 0)$ is a minimal nilsystem by Leibman's orbit-closure theorem [4]. Consequently X projects onto that closure, and its two coordinates factor through Z:

$$\pi : X \to Z, \qquad \rho : Z \to Y_0, \qquad \gamma : Z \to \mathbb{T}^M,$$

where Y_0 is the orbit closure of y and $\gamma(Tz) = \gamma(z) + \alpha$. Put $W = \rho^{-1}(V \cap Y_0)$. Let $C_1 \subseteq X$ denote the clopen cylinder of primary color 1, and let L_i denote the clopen cylinder of auxiliary color i.

Case 1: $W \subseteq \pi(C_1)$**.** For each $i \in [r]$ put $F_i = \pi(C_1 \cap L_i)$. These sets are closed and their union covers W. The boundary of a closed set is nowhere dense. By the Baire category theorem, choose $z_0 \in W$ whose entire orbit avoids all the boundaries ∂F_i: we remove the countable union of all their integer translates. Thus each $T^n z_0 \in W$ belongs to the interior of at least one F_i. This makes it possible to specify an auxiliary color and still use an *open* intersection in Lemma 4.1.

Apply Lemma 3.1 with $y' = \rho(z_0) \in V$ in the original nilsystem, with its original dimension bound C. It supplies vertex times n_v whose phases have the prescribed edge differences, with $T^{n_v} y' \in V$ and all required orbit segments lying in V. Give vertex v any label i for which $T^{n_v} z_0 \in \operatorname{Int} F_i$. The graph has chromatic number greater than r, so there is a directed edge $u \to v$ on which the labels agree, say with value i. Write

$$m = n_v - n_u, \qquad z_* = T^{n_u} z_0, \qquad z_j = T^{jm} z_* \quad (0 \leq j < h).$$

All z_j lie in W, the first two belong to $\operatorname{Int} F_i$, and $m\alpha \in O_M$.

Choose open phase neighborhoods P_0 of $\gamma(z_*)$ and P_1 of $\gamma(z_*) + m\alpha$ so that $P_1 - P_0 \subseteq O_M$. For $j = 0, 1$ define

$$U_j = (C_1 \cap L_i) \cap \pi^{-1}\big(W \cap \gamma^{-1}(P_j)\big).$$

For $2 \leq j < h$, put $U_j = C_1 \cap \pi^{-1}(W)$. For $j = 0, 1$ the image $\pi(U_j)$ contains the open neighborhood $\operatorname{Int} F_i \cap W \cap \gamma^{-1}(P_j)$ of z_j. For the other j the image is all of W. The points z_j therefore verify the hypothesis of Lemma 4.1. It gives an h-term configuration in these sets with a positive step $d = m + n^2$. Its primary colors all equal 1, its first two auxiliary colors both equal i, and all its Y-coordinates are in V. The first two torus phases differ by $d\alpha \in P_1 - P_0 \subseteq O_M$, so $d \in G_M$. The preliminary edge difference m and the final difference d need not be the same.

Case 2: $W \not\subseteq \pi(C_1)$**.** Here $W \setminus \pi(C_1)$ is nonempty and open. By the inverse-limit topology, it contains $\zeta^{-1}(V')$ for a finite-dimensional minimal nilsystem factor $\zeta : Z \to Z'$ and a nonempty open $V' \subseteq Z'$. Choose $x' \in X$ with $z' = \zeta\pi(x') \in V'$. Choose an open neighborhood O of zero in $\mathbb{T}^M$ so small that every vector $v \in O - O$ satisfies $\sum_{i \leq M} \|v_i\|_{\mathbb{R}/\mathbb{Z}} < \tau$. Use the point $(z', 0)$ and the window $V' \times O$ in $(Z' \times \mathbb{T}^M, T \times R_\alpha)$.

At visits to this window use the primary color of $T^n x'$. There are at most s such colors because no point above V' belongs to C_1. For all $n \geq 1$, read the auxiliary color at the same point $T^n x'$. Applying W_s of B gives an h-term progression of visits with common primary color, equal first

two auxiliary colors, and step $d \in B$. Its first two phase coordinates lie in O, so $d \in B_M$. The corresponding points along the orbit of x' all project into V. For $s = 0$, this second case cannot occur because $C_1 = X$.

Transfer. In either case, the configuration in X specifies finitely many symbolic coordinates and finitely many visits to the open window V. These conditions are open for the joined system and hold at the starting point of the configuration. Since X lies in the forward limit set, such a configuration occurs at a positive starting time in the original joined orbit, with the same d. Every primary color used there is a color of the original return-time set, and the auxiliary equality transfers at the same two sites. □

5 A coloring for the infinite set of steps

Retain the constants in (2.1), and now let $\alpha_1, \alpha_2, \dots$ be a positive summable sequence with $1, \alpha_1, \alpha_2, \dots$ rationally independent. Let G_∞ consist of the $d \in \mathbb{N}$ for which the centered coordinates θ_j of $(d\alpha_j)_{j\ge1}$ satisfy the same conditions on the whole sequence: there is an $i \ge 1$ with

$$|\theta_i + \delta| < \eta, \qquad |\theta_j| < \eta \quad (j \ne i), \qquad \sum_{j\ne i} |\theta_j| < 2\delta.$$

The distinguished index i is unique because $2\eta < \delta$. For $B \subseteq \mathbb{N}$ and an integer $H \ge 2$, put

$$\tau = (4\pi qH)^{-1}, \qquad B_\infty = \left\{ d \in B : \sum_{j=1}^{\infty} \|d\alpha_j\|_{\mathbb{R}/\mathbb{Z}} < \tau \right\}.$$

The idea of detecting one distinguished coordinate by the curvature of a sum of circle phases comes from Alweiss [1, Section 2]. Here we use the real part of that sum to select which of q output colors is missing at each integer.

Lemma 5.1. *Suppose $f : \mathbb{N} \to \{0, \dots, u-1\}$ has no monochromatic H-term progression with step in B. Then there is a coloring $\kappa : \mathbb{N} \to \mathbb{Z}/q\mathbb{Z}$ with no monochromatic progression of length $\max\{200q, H\}$ with step in $G_\infty \cup B_\infty$.*

Proof. For every integer J, we define an injection $\iota_J : \{0, \dots, u-1\} \to \mathbb{Z}/q\mathbb{Z}$. If the input is i, let t be the unique integer in $[J-u, J-1]$ congruent to i modulo u, and set $\iota_J(i) = t - 1 \pmod q$. Its image consists of the residues represented by $J-q, \dots, J-2$, so its missing output is $J-1 \pmod q$. When J is replaced by $J+1$, the representative $J-u$ leaves the interval and J enters it. These represent the same input, and its output changes to the previously missing color; all other inputs keep their outputs. It follows that

$$|J - J'| \le 1, \quad \iota_J(i) = \iota_{J'}(i') \qquad \Longrightarrow \qquad i = i'.$$

Write $\mathrm{e}(t) = \exp(2\pi i t)$, and define

$$F(n) = \sum_{j=1}^{\infty} \mathrm{Re}(\mathrm{e}(n\alpha_j) - 1), \qquad J(n) = \lfloor qF(n) \rfloor, \qquad \kappa(n) = \iota_{J(n)}(f(n)).$$

The series converges absolutely for each n, since $|\mathrm{e}(n\alpha_j) - 1| \le 2\pi \|n\alpha_j\|_{\mathbb{R}/\mathbb{Z}} \le 2\pi n\alpha_j$ and $\sum_j \alpha_j < \infty$.

First suppose $d \in B_\infty$. For any $0 \le r, t < H$ and $b \in \mathbb{N}$, the phase increment bound gives

$$|qF(b+td) - qF(b+rd)| \le 2\pi q|t-r| \sum_{j=1}^{\infty} \|d\alpha_j\|_{\mathbb{R}/\mathbb{Z}} < \tfrac{1}{2}.$$

Thus all H values of J lie in at most two consecutive integers. If the H outputs of κ were constant, the compatibility of these injections would force all H inputs from f to be equal, contrary to $d \in B$.

Now suppose $d \in G_\infty$, with distinguished coordinate $\theta_i = -1/Q + \epsilon$, where $|\epsilon| < \eta$. Fix $b \in \mathbb{N}$ and write $F_r = F(b+rd)$ for $0 \le r < 2Q$. Consecutive samples satisfy

$$|qF_{r+1} - qF_r| \le 2\pi q \sum_{j \ge 1} |\theta_j| < 2\pi q(3/Q + \eta) < 1,$$

so the floors of qF_r change by at most one at each move. We show that their range is large enough to visit every residue modulo q.

Choose $0 \le t < Q$ with $\|b\alpha_i - t/Q\|_{\mathbb{R}/\mathbb{Z}} \le 1/(2Q)$, using the grid of Q equally spaced points on the circle. Then $\|b\alpha_i + t\theta_i\|_{\mathbb{R}/\mathbb{Z}} \le 1/(2Q) + Q\eta < 1/Q$, so $z = \mathrm{e}(b\alpha_i + t\theta_i)$ has real part greater than 0.99 because $Q \ge 200$. Put $m = Q/2$, and note that $t, t+m, t+2m$ all belong to $\{0, \dots, 2Q-1\}$. The contribution of the distinguished coordinate to $F_{t+2m} - 2F_{t+m} + F_t$ is

$$\mathrm{Re}(z(\mathrm{e}(m\theta_i) - 1)^2).$$

Since $m\theta_i = -1/2 + m\epsilon$, we can write $\mathrm{e}(m\theta_i) = -1 + w$ with $|w| \le \pi Q\eta$. Expanding $(-2+w)^2$ shows that the distinguished contribution is greater than $3.96 - 4\pi Q\eta - \pi^2 Q^2\eta^2$. The absolute sum of the contributions from *all* other coordinates is at most

$$\sum_{j \ne i} |\mathrm{e}(m\theta_j) - 1|^2 \le \pi^2 Q^2 \sum_{j \ne i} |\theta_j|^2 \le \pi^2 Q^2 \eta \sum_{j \ne i} |\theta_j| < 2\pi^2 Q\eta.$$

In particular, this bound includes the entire infinite tail. As $Q\eta = 1/(10^6 Q)$, the full second difference exceeds 3.9. It is at most twice the range of the $2Q$ values F_r, so that range exceeds 1.95. The range of the qF_r therefore exceeds q, and their floors, which move by at most one, visit at least q consecutive integers. For each color $c \in \mathbb{Z}/q\mathbb{Z}$, some visited J satisfies $J - 1 \equiv c \pmod q$, so that the corresponding injection omits c. Consequently the $2Q = 200q$ sites cannot all have the same output color. Together with the first case, this proves the lemma. □

6 One summable sequence of frequencies

We now pass from Proposition 4.2 to one set of integers that works for every test in W_{s+1}. Fix the old set B, its blocking coloring f and its forbidden length H for this entire induction step. As before, $u = s+1$, $q = s+2$, $Q = 100q$, $\delta = 1/Q$, $\eta = 1/(10^6 Q^2)$, and $\tau = 1/(4\pi qH)$. We choose increasingly long initial segments of a single sequence of positive frequencies $(\alpha_i)_{i \ge 1}$. The rapid decrease of the frequencies plays the same role as in finite-cutoff constructions for nil-Bohr recurrence; compare Green [3]. Here the cutoff at each stage is determined by finitely many steps that suffice for a whole family of colorings and niltranslations.

A countable list of compact tests

We first explain how to list the nilsystem tests without changing their integer time parameter. Given a niltranslation on a compact quotient $Y = G_0/\Gamma_0$, we may restrict the acting group to the

subgroup generated by its identity component, representatives of the finitely many components of Y, and the translating element. This subgroup still acts transitively on Y and has finitely generated component group. By Leibman's embedding of nilmanifolds [4, §§ 1.11 and 2.1], there is then an embedding of Y into a nilmanifold of a connected simply connected nilpotent Lie group under which the original translation is represented by an ambient translation. If the ambient uniform subgroup is not discrete, quotienting by its identity component gives a lattice presentation of the same ambient space [4, § 2.8]. The quotient group remains connected and simply connected, since the normal subgroup removed is connected. Thus we may use an ambient model G/Γ with G connected and simply connected and Γ a lattice. The dimension of this model may exceed $\dim Y$, but the model is fixed before the finite-stage proposition is applied.

If the image of the original starting point is $z\Gamma$, left translation by z^{-1} moves it to $o = e\Gamma$ and conjugates the ambient translation to left translation by some $b \in G$. For an original open window $U \ni y$, we may choose an ambient neighborhood of o whose intersection with the translated embedded copy of Y is contained in the transported U. Consequently, visits of $b^n o$ to this ambient neighborhood give visits of $T^n y$ to U at the very same positive times n.

There are only countably many pairs (G, Γ) of this kind up to isomorphism: by Mal'cev's rationality theorem, rational Lie structure constants and the rational logarithmic coordinates of finitely many lattice generators describe such a pair [5]. For each representative choose a compact exhaustion of G by nonempty sets and a countable neighborhood basis at o in G/Γ. List all tests

$$(G, \Gamma, K, V, h, r),$$

where K belongs to that exhaustion, V belongs to that basis, $h \geq 2$, and $r \geq 1$. Here b ranges over K, the primary palette has $s+1$ colors, and the auxiliary palette has r colors. Every original test for W_{s+1} is captured by some entry: take a compact set containing b and a basis neighborhood inside the ambient neighborhood just described. The primary coloring, originally specified only on the relevant visiting times, can be extended arbitrarily to all positive times.

Finitely many steps at each stage

Set $M_0 = 0$. At stage j, take the jth entry of the list and choose $M_j > M_{j-1}$ large enough for Proposition 4.2 with its parameters $h, \dim G, q, r$. Retain all previously chosen frequencies and choose the new positive frequencies so that $1, \alpha_1, \dots, \alpha_{M_j}$ are rationally independent. Their total size may be required to be arbitrarily small: at each selection only the countable rational span of the preceding numbers must be avoided. The precise upper bound imposed by the preceding stages is given below. Write $G_{M_j}, B_{M_j}, S_{M_j}$ for the sets formed from this prefix.

Consider the compact parameter space

$$K \times \{1, \dots, s+1\}^{\mathbb{N}} \times \{1, \dots, r\}^{\mathbb{N}}.$$

A parameter (b, c, g) specifies the translation and the two colorings. For fixed $a \geq 1$ and $d \in S_{M_j}$, the conditions

$$\begin{aligned} &c(a) = c(a+d) = \cdots = c(a+(h-1)d), && g(a) = g(a+d), \\ &b^{a+td} o \in V && (0 \leq t < h) \end{aligned}$$

define an open set in this parameter space. Indeed, the color equalities inspect finitely many coordinates and each displayed iterate is continuous in b. These open sets cover the parameter space by Proposition 4.2. Choose a finite subcover and let F_j be the finite nonempty set of its differences. For every parameter in this test, at least one $d \in F_j$ works with a positive start a.

Preserving all previously selected steps

For each $d \in F_j$, record one reason for its membership in $S_{M_j} = G_{M_j} \cup B_{M_j}$. Write $\|x\|$ for distance to the nearest integer. If the recorded reason is G_{M_j}, fix a distinguished index $i = i(d) \le M_j$ and set

$$\sigma_d = \min\left\{ \eta,\ 2\delta - \sum_{\substack{t \le M_j \\ t \ne i}} \|d\alpha_t\| \right\}.$$

The two old coordinatewise conditions are already satisfied, and $\sigma_d > 0$ by the strict inequality on the sum. If the recorded reason is B_{M_j}, set

$$\sigma_d = \tau - \sum_{t \le M_j} \|d\alpha_t\| > 0.$$

Choose $t_j > 0$ so small that

$$(\max F_j) t_j < \min_{d \in F_j} \sigma_d.$$

After stage ℓ impose on the next block the requirement

$$\sum_{M_\ell < i \le M_{\ell+1}} \alpha_i < 2^{-\ell} \min\{1, t_1, \dots, t_\ell\}.$$

There is no restriction from an earlier block when selecting the first one. The procedure is sequential: the prefix and the finite subcover are chosen before the new tolerance t_j, and t_j constrains only frequencies still to be selected. It follows that $\sum_i \alpha_i < \infty$ and, for each j,

$$\sum_{i > M_j} \alpha_i \le t_j \sum_{\ell=j}^{\infty} 2^{-\ell} \le t_j.$$

Define G_∞ by the same distinguished-coordinate conditions as G_M, now over all coordinates, with the sum of the norms of the remaining coordinates strictly below 2δ. Define

$$B_\infty = \left\{ d \in B : \sum_{i=1}^{\infty} \|d\alpha_i\| < \tau \right\}, \qquad D_{s+1} = G_\infty \cup B_\infty.$$

For $d \in F_j$ the total contribution of all new coordinates is at most

$$\sum_{i > M_j} \|d\alpha_i\| \le d \sum_{i > M_j} \alpha_i \le (\max F_j) t_j < \sigma_d.$$

A recorded G_{M_j}-step therefore keeps its distinguished coordinate, every new coordinate has norm less than η, and the sum over the remaining coordinates stays below 2δ. A recorded B_{M_j}-step remains in B and its total norm stays below τ. In particular, $F_j \subseteq D_{s+1}$ for every j.

The resulting recurrence and obstruction

Every test has its own stage, and the finite set F_j still supplies the required progression and the auxiliary equality on its first two terms. Transporting from the ambient model back to the original window proves W_{s+1} for D_{s+1}. For the upper bound, summability makes the global series

$$\sum_{i=1}^{\infty} \mathrm{Re}\big(\exp(2\pi i n \alpha_i) - 1\big)$$

absolutely convergent at every n, since the absolute value of its ith term is at most $2\pi n\alpha_i$. Apply Lemma 5.1 to the full frequency sequence and the fixed old coloring f. It gives a single q-coloring blocking all D_{s+1}-progressions of length $\max\{2Q, H\}$. Only membership of the selected differences had to survive the addition of new coordinates; the global blocking coloring is obtained directly from the full series.

For the initial step use $s = 0$, $B = \varnothing$, $H_0 = 2$, and the constant input coloring, as in Proposition 4.2. Repeating the construction with the resulting old set and coloring gives D_k with W_k and a $(k+1)$-color obstruction for every $k \geq 1$. The forbidden lengths satisfy

$$H_0 = 2, \qquad H_k = \max\{200(k+1), H_{k-1}\} = 200(k+1) \quad (k \geq 1).$$

7 Consequences for the original question

The one-point instance of W_k proves that $D_k \cap A$ is k-large for every $A \in \mathcal{P}^*$, by the finite-cover argument of Corollary 1.3. In particular, for $k = 2$ the counterexample satisfies all the requested restrictions simultaneously.

In fact $D_k \cap A$ retains W_k itself. Given an auxiliary coloring g, use the product of g and the finite-cover coloring for A as the auxiliary coloring in W_k for D_k. Equality of the first two product colors ensures both the requested equality for g and membership of the step in A. The obstructing coloring for D_k still blocks progressions with steps in any subset, so every one of these restrictions is itself k-large but not $(k+1)$-large.

One can also insist that the counterexample to largeness itself be syndetic. Put $S = D_2 \cup (2\mathbb{N}-1)$. Every $S \cap A$ contains $D_2 \cap A$ and hence is 2-large. If κ is the obstructing three-coloring for D_2, the six-coloring $n \mapsto (n \bmod 2, \kappa(n))$ blocks all 600-term S-progressions. Odd differences change parity; an even difference in S belongs to D_2. This last argument proves nonlargeness for the syndetic enlargement; the three-color bound applies to D_2 itself.

The proof uses no hypothesis identifying Bohr recurrence with topological recurrence. The equal-color pair is obtained for an arbitrary finite coloring by the finite shift graph; imposing only finitely many torus constraints in place of that coloring would not justify the conclusion.

8 A shorter route if only nonlargeness is required

There is a second proof of the negative answer that stops after the one-color case of the construction. It gives a syndetic counterexample without the three-color obstruction proved for D_2. This route uses the exceptional-set theorem of Glasscock, Koutsogiannis, Le, Moreira, Richter, and Robertson [2, Theorem A].

Take $E = D_1$. To construct E alone, Lemma 3.1 proves the finite-stage recurrence directly: label vertex v by the actual auxiliary color $g(n_v)$ and take a monochromatic edge. There is only one primary color. Sections 5 and 6 then give E with W_1 and a binary obstruction of length 400, without using Qiu or a color induction.

Fix an irrational θ and put

$$F = \{d \geq 1 : \|\theta d^2\|_{\mathbb{R}/\mathbb{Z}} > 1/4\}, \qquad S = E \cup F.$$

A finite coloring blocks all three-term progressions of difference in F: color n by the interval of length $1/16$ containing the fractional part of $\theta n^2/2$. For three points in one interval the absolute value of their second difference is at most $2/16$. Modulo one the second difference here equals θd^2. The product of this coloring and the binary obstruction for E proves that S is nonlarge.

Why $F \cap A$ is piecewise syndetic for every $A \in \mathcal{P}^*$. Choose a finite coloring g such that any equal-color pair has its difference in A, as in Corollary 1.3. Write r for its number of colors and set $I = (0.24, 0.26) \subset \mathbb{T}$. For each $\epsilon > 0$ the set of n with $\|n\theta\|_{\mathbb{R}/\mathbb{Z}} < \epsilon$ and $\{n^2\theta\} \in I$ is syndetic. Indeed it is an open-set visiting set for the point $(0, 0)$ under the minimal skew shift

$$(x, y) \longmapsto (x + \theta,\ y + 2x + \theta)$$

of $\mathbb{T}^2$. Every orbit is dense by Weyl's criterion: a character involving the second coordinate has irrational quadratic coefficient, and any nonconstant character using only the first has irrational linear coefficient.

Choose successively $r + 1$ centers $n_1 < \cdots < n_{r+1}$. At stage j, use the syndetic set of n with $n > n_{j-1}$ (omit this bound at $j = 1$), $\{n^2\theta\} \in I$, and $\|n\theta\|_{\mathbb{R}/\mathbb{Z}}$ so small that $\|2n_i n\theta\|_{\mathbb{R}/\mathbb{Z}} < 0.01$ for all $i < j$. Since piecewise syndeticity is partition regular, some g-color class P_j in this candidate set is piecewise syndetic. Choose $n_j \in P_j$ and record its color. Two recorded colors coincide, say at $i < j$. For every $n \in P_j$, the positive difference $d = n - n_i$ belongs to A, while

$$\theta d^2 = \theta n^2 + \theta n_i^2 - 2n_i n\theta$$

lies modulo one between 0.47 and 0.53, and hence $d \in F$. Thus $F \cap A$ contains the piecewise syndetic set $P_j - n_i$. The set F itself is syndetic as an open-set visiting set of the same skew shift, so S is syndetic.

Why $S \cap A$ is 2-large. We already know that $E \cap A$ has W_1, by multiplying auxiliary colorings as in the preceding section. Take a minimal binary subshift in the forward limit set of the given coloring, and let $\pi : X \to Z$ be its maximal infinite-step pronilfactor. Write C_0, C_1 for the two color cylinders and fix $h \geq 2$. If $\pi(C_0) = Z$, Theorem A of [2], applied to $0, n, \ldots, (h-1)n$ and h copies of C_0, implies that the complement of the h-term progression-return set for C_0 is not piecewise syndetic. It cannot contain $F \cap A$, so there is a suitable step in $F \cap A$. If $\pi(C_0) \neq Z$, take a finite nilfactor and a nonempty open window whose whole preimage in Z is contained in $Z \setminus \pi(C_0)$. Property W_1 of $E \cap A$, with constant auxiliary coloring and any starting point in that window, supplies a progression of visits with step in $E \cap A$. Lifting the start to X gives a progression inside C_1. The usual forward-limit transfer returns either progression to the original coloring. This gives a direct proof for the proposed hypothesis using only one shift-graph construction.

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